Adrian Erlikhman

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Education

Los Angeles Center for Enriched Studies (LACES) Class of 2027

LAUSD magnet school | Los Angeles, CA

- GPA 4.0 unweighted, 5.2 weighted | SAT 1520 (Math 770, Reading and Writing 750)
- 17 AP courses (11 completed, 6 in progress) across computer science, mathematics and the sciences.

Los Angeles Southwest College

Dual enrollment | Los Angeles Community College District

- Linear Algebra (MATH 270) | Programming in Python (CS 119) | Python Programming for Data Science and Machine Learning (CS 121, planned)

Research Interests

Evaluation of language models and of the benchmarks used to rank them: authorship attribution and stylometry, the reliability of LLM judges, and reward-model and judge selection. Robustness of standard quantitative methods in portfolio construction and composite indices.

Publications

Conference and workshop papers

[1] **A. Erlikhman**. "Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation." *First Workshop on Autonomous Research and Recursive Self-Improvement (ARRSI), AACL-IJCNLP 2026*, Hengqin, China, Nov 2026. **Accepted.** Sole author. PDF Code

- Introduced curation certificates: how much of a validation set could be removed before a different reward model or LLM judge wins. Measured in benchmark weight, the certificate is at least the top-two score gap; measured in items, it is computed exactly.
- Removing 12 particular items of RewardBench's 2,985 makes the runner-up the winner (11 under RewardBench's own scoring code). On random 200-item validation sets, the 8% of selections with a top-two gap above 2 points match the full-benchmark winner 87% of the time (43% otherwise).

[2] **A. Erlikhman**, M. Tarekegn, P. Juang. "Interpretable Multilingual Attribution of Frontier LLMs." *IEEE BigData 2026 High School Symposium*, Phoenix, AZ, Dec 2026. **Accepted; Best Paper Finalist.** Co-first author. PDF Code

- Attributed 840 essays from five frontier models in seven languages, English to Hindi, with 21 interpretable features (the syntactic ones from Universal Dependencies parses): 59–72% accuracy in every language (chance 20%), with no decline detected as language resources fall.
- Character n-grams are 3–22 points more accurate within a language, but the features transfer between languages better (49% vs. 41%) and reach 57–70% on machine-translated text; across 4,195 judge calls, four of the five models attribute authorship at chance.

Journal articles

[3] **A. Erlikhman**, R. Erlikhman. "Does Portfolio Optimization Beat Splitting Evenly?" *Oxford Journal of Student Scholarship* 12(7):510–529, Oct 2026. **Published.** Co-first author. PDF

- Equal weights beat Markowitz in 74% of 250 calibrated simulations (Sharpe ratio 0.70 vs. 0.51). Markowitz's shortfall comes from estimating expected returns; shrinkage overtakes equal weights only with roughly two decades of stationary data.

Manuscripts in revision and preprints

[4] **A. Erlikhman**, R. Erlikhman. "How Robust Are the Legatum Prosperity Index Rankings? A Monte Carlo and Sensitivity Analysis of Weighting, Normalization, and Aggregation." *Journal of High School Science*, 2026. **In revision.** Co-first author. PDF

- Reproduced every published score for 167 countries, then ran 10,000 draws under each of three weight priors and a 131,072-run Sobol' design over weighting, normalization and aggregation.
- Derived rank leverage, $c^2(1-\rho^2)$, a closed-form measure of each pillar's potential to change ranks, which tracks Sobol' total effects (Spearman 0.91–0.99). The ranking holds as a whole (median Spearman ≥ 0.9899) but not for ranks 68–100, where the median 90% rank interval spans 36 places.

[5] **A. Erlikhman**, M. Tarekegn, P. Juang. "Whose Line Is It Anyway? Auditing Self- and Other-Recognition in LLM Judges." Preprint, arXiv forthcoming, 2026. Co-first author. Code and data

- Introduced a peer baseline that scores each LLM judge's self-attribution against how often the other judges name it on the same text, separating self-recognition from text any model can identify.
- Nine judges, 1,270 texts, over 19,000 judgments: the standard self-naming test flags 15 of 37 cases, 8 survive a false-alarm control, 3 significantly beat the pooled peers, and none significantly beats its strongest single peer. An 18-feature stylometric classifier attributes 86.3% of the chat-app essays, more than any judge.

Presentations

"Interpretable Multilingual Attribution of Frontier LLMs." *IEEE BigData 2026 High School Symposium*, Phoenix, AZ (scheduled). — Dec 2026

Invited panelist, student panel on AI, in aiEDU's pre-conference session "Transform Your School's AI Readiness." *National Consortium of STEM Schools (NCSS) 2026*, San Diego, CA (scheduled). — Dec 2026

"Curated to Win: Certifying Reward-Model and Judge Selection Against Evaluation-Set Curation." *ARRSI Workshop, AACL-IJCNLP 2026*, virtual (scheduled). — Nov 2026

Invited guest, aiEDU podcast, hosted by CEO Alex Kotran. aiEDU is a national AI-literacy nonprofit. — TBA

Presenter, the AIML-LI curriculum. *LAUSD Innovation Expo*, Los Angeles.

Honors & Awards

Best Paper Finalist, IEEE BigData 2026 High School Symposium (one of 12 finalists among 130 accepted papers) — 2026

National Merit Semifinalist (about 16,000 of 1.3M+ entrants) — 2026

1st place, Council District 9 Champion (U18), VISION HACK: South LA, for Ledger — 2026

1st place, Finance track, ACM TrojanHacks, for Monte — 2026

Winner, Agent Behaviour Monitoring track, SEPHacks, judged by Lemma, for Hivemind — 2026

1st place, Decode the Ocean Hackathon (Lovable × United Nations), for eDNAtlas — 2026

3rd place internationally, Code for Transportation (Young Coders' Sphere), for Safe Routes to School — 2026

Finalist, top 8 of 45 teams, Citadel High School Terminal — 2026

Admitted, Y Combinator Startup School, as AIML-LI's co-founder — 2026

Teen Innovation Grant, Jewish Federation of Greater Los Angeles (JFEDLA), for SafeJew — 2024–2025

2× AP Scholar with Distinction — 2025, 2026

USA Fencing: 3× All-American and 3× All-Academic, both First Team

SIFMA Stock Market Game: top 5% statewide, as a solo entrant

Music Teachers' Association of California (MTAC), piano: five state-honors exams; Level 9 candidate

Experience

Firstness — 2026 – Present

Software Engineering & ML Intern | firstness.ai, an AI research engine | reporting directly to the founder

- Firstness grows a research question into a map of branching hypotheses, writes each answer from the peer-reviewed literature with a citation on every sentence, and checks a new hypothesis against published prior work.
- Refine how the platform connects into the research stack: the literature databases and reference managers researchers already use.
- Build demos that show new users what the engine does and how to check each answer against its cited source.

Kiddom — 2026 – Present

Machine Learning Internship & Mentorship | kiddom.co | with Director of Data Science Flora Xu

- Design ML and NLP approaches to teacher-facing problems in K–12 assessment: which problems merit modeling, what student data schools can share, and what result counts as evidence.

- Extending **Why Wrong**, built independently, as the mentorship's project. It reads a class's answer sheets, names the misconception behind each wrong answer (naive Bayes over answer patterns, then clustering on the residual errors) and groups the class by shared mistake. F1 0.87 on planted misconceptions (1.0 is perfect). Live Code
- Design its classroom evaluation with an LAUSD teacher: a review of the item bank and misconception labels, a pilot on one class's de-identified responses, and a write-up of where tool and teacher agree and disagree.

Fjor Venture Capital 2024 – 2026

Investment & Software Engineering Intern | $15M early-stage fund

- Built classifier and scoring models to screen 134 startups in Y Combinator's Summer 2025 batch, and led the screen with Managing Director Tatiana Janko, from sourcing and first-pass review to a competitive landscape for each shortlisted company.
- Ran diligence on 20+ early-stage AI startups across enterprise SaaS, fintech and deep tech, and wrote 10+ investment memos for partner decisions.
- Built the fund's internal software tools, and drafted its monthly limited-partner (LP) updates and investor-facing materials.

Alliance for SoCal Innovation 2024 – 2026

Venture & Data Strategy Intern | nonprofit coalition of USC, UCLA, Caltech and 20+ organizations

- Built a Python/SQL engine that matches startups to likely investors by investment history and company characteristics, over 10,000 contacts at 250+ VC firms, angel syndicates and accelerators; deduplication and standardization cut founder-to-investor lookups from days to under a minute.
- Worked with Southern California's deep-tech startups, including companies spun out of Caltech: diligenced 35 startups, led outreach to 80+ founders and mapped the region's deep-tech ecosystem.
- Led the venture pipeline's backend redesign with the Chief Innovation Officer.

Advanced Mathematics 2022 – Present

With Kirill Kovalenko, PhD candidate | Induction, number theory, combinatorics and game theory

Selected Projects

AIML-LI (AI/ML Literacy Initiative) 2023 – Present

Co-Founder & Lead | aiml-initiative.org

- Co-wrote a free AI-literacy course: twelve 30-minute sessions that run on classroom Chromebooks, plus a 12-week, six-unit version with Colab notebooks, instructor guides, student packets and assessments. Taught in LACES AP Computer Science, spring 2026.
- Took it to LAUSD's Ed Tech & Innovation office (Sep 2026), which is carrying it to the district's Chief Academic and Chief Information Officers toward a standing AI-literacy elective. Advised by Philo Juang (Google DeepMind) and Amir Zohrenejad (Data Herald, Y Combinator W21); mentored by aiEDU.
- Organize the LA Student AI Summit & Hackathon (Feb 2027) for 500+ LA public-school students, with keynotes from Google DeepMind, Microsoft, Google and Snap.

SafeJew 2024 – Present

Co-Founder | safejew.org

- Built a live map of antisemitic incidents: 800 across Greater LA (100 individually sourced, each with a citable link) and 594 on 33 U.S. university campuses. Wrote a dependency-free LA gazetteer that geocodes places named in news articles, which lets a daily GitHub Action update the map unattended.
- Funded by a Teen Innovation Grant from the Jewish Federation of Greater Los Angeles (JFEDLA) and distributed through its Community Security Initiative.

Safe Routes to School 2026

Co-Creator | 3rd place internationally, Code for Transportation | safe-routes-la.github.io

- Scores 431,599 LA street blocks for street-crime risk from 85,634 filtered LAPD incidents, then runs A* over them in the browser across the 668 LAUSD schools. On one modeled trip, Exposition & Normandie to LACES, two buses cut modeled street-crime exposure by 80% and take 40 minutes against 118 on foot.

Ledger Sep 2026

1st place, VISION HACK: South LA | Hacker Fund × City of LA Vision Lab

- Puts the 18 programs a corner store answers to on one renewal calendar, and scores a photo of a wholesale order record against the SNAP stocking rule. Led the build across the vision pipeline, rule engine and frontend, and wrote the two-pass vision pipeline, the dashboard and renewal calendar, and the static fallback. Live Code

Hivemind Oct 2026

Track winner, SEPHacks 2026 | with Michael Tarekegn

- Shared memory for a team's AI coding agents: an engineer's correction becomes a lesson, the team lead approves it, and every agent loads it at its next session. Built the website, the web app and the terminal app's design, and made the launch film; Michael built the engine and eval. Claude Code passed 36 of 36 eval runs with it, 14 of 36 without, and 15 of 36 with a placebo. Live Devpost

Monte Oct 2026

1st place, Finance track, ACM TrojanHacks 2026 | with Michael Tarekegn

- Picks the credit card that pays most at each purchase, never one carrying a balance. Built the interface, the tap-to-pay demo and the brand, and made the launch film; Michael built the backend, the Monte Carlo and the tests. Live Devpost

dunderscore 2026

Finalist, Citadel High School Terminal | with two teammates

- Co-wrote a Python bot for Terminal, Citadel's algorithmic tower-defense game: it reads the board, prices every wall, turret and attack lane, and adapts to the opponent's last layout; revised after each round's replay review.

eDNAtlas 2026

1st place, Decode the Ocean Hackathon | Lovable × United Nations

- Built a live map that turns environmental DNA sampling into a plain-language health score for coastal sites, with the raw evidence and downloads in Darwin Core, the standard biodiversity-data format, behind every score. Live Code

Leadership & Service

Jewish Student Union 2024 – Present

President (Secretary, 2024–2025) | 50+ members

- Run weekly programming with NCSY, a Jewish youth movement; organized a school Holocaust remembrance assembly with a survivor and StandWithUs.

STEMsters 2022 – Present

Co-Founder & President | 30+ volunteers

- Place LACES volunteers in four elementary schools to teach hands-on coding, robotics and engineering.

LA International Fencing Center

Coach

- Coach beginner classes for younger kids, and coach my younger brother at the club and at tournaments.

Athletics

Competitive Fencing, Épée

USA Fencing A26 (the highest rating), Team USA | LA International Fencing Center

- As of Nov 2026: U.S. No. 49 Junior (U20) and No. 78 Senior; peak No. 17 Cadet (U17); No. 1 in Region 4. Represented Team USA at the 2026 Cadet Pan-American Championships in Bogotá (6th). In active recruitment with MIT.
- Top 32 Division I, National Championships | Top 8 Cadet, Super Junior and Cadet Circuit (SJCC) | Top 16 Cadet, October North American Cup | 4× Junior (U20) regional champion

Skills

Code: Python, TypeScript/JavaScript, Java, SQL, LaTeX; Next.js, React, Supabase/PostgreSQL, GitHub Actions, Git

ML & statistics: PyTorch, scikit-learn, TensorFlow/Keras, pandas, NumPy, SHAP, Stanza, FinBERT; LLM evaluation, stylometry, Monte Carlo and Sobol' sensitivity analysis, hypothesis testing, time series, HMMs

Certifications: Certiport IT Specialist in Software Development, Cybersecurity, Python, Data Analytics, AI and HTML/CSS; Deep Learning Specialization (Coursera); NVIDIA Deep Learning Fundamentals; Kaggle × Google Generative AI Intensive

Languages: English, Russian (fluent), Spanish (intermediate)